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AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:
US/Europe Divergence Trajectories 2024–2030

Integrated Geostrategic and Economic Analysis

Chapter IV

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77% global AI compute = USA **1.6×** energy cost EU/US
3.4:1 CACI ratio US/EU (Power Mode)

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CHAPTER IV

Mechanisms of American Competitive Advantage Through AI

Chapter III established the facts: the United States concentrates 75% of global AI compute, 45% of data center energy consumption, and controls 70% of the European cloud market through three hyperscalers. This chapter analyzes the *mechanisms* through which this compute asymmetry translates into measurable competitive advantage. We identify four transmission channels: training costs as a barrier to entry, cloud concentration as a vector of dependence, regionally differentiated productivity, and the capture of innovation rents by first movers.

4.1 Training Costs as a Barrier to Entry

4.1.1 The Exponential Explosion of Training Costs

The most documented channel is that of foundation model training costs. Martens (Bruegel, October 2024) establishes that the cost of training a frontier model rose from **\$1,000 in 2017 to nearly \$200 million in 2024**, and could reach **several billion by 2030**.¹ This exponential growth follows the *scaling laws* identified by Kaplan et al. (2020): the improvement of cognitive performance in models is subject to constant returns in complementary inputs (parameters, training data, compute capacity). In other words, each marginal gain in performance requires a proportional increase in compute, data, and model size.

Cottier et al. (2024), cited by Martens, decompose training costs into five components: personnel, AI chips, server costs, network interconnection, and energy. They estimate infrastructure costs (hardware + data center) at **ten times the cost of training itself**, with a depreciation rate of 140%/year (complete depreciation in 8.5 months, matching the renewal pace of chip generations). GPT-4 infrastructure by end-2023 was estimated at approximately \$800 million. By extrapolation, infrastructure costs could reach \$500 billion by 2030, replicated across half a dozen hyperscalers.²

4.1.2 The US Advantage: Abundant and Subsidized Compute

In this context, access to cutting-edge compute becomes the discriminating factor. American Big Tech (Microsoft, Meta, Google, Amazon, xAI) collectively invested **\$320 billion in 2025** in AI infrastructure (servers, data centers, chips), up from \$230 billion in 2024.³ Meta is developing an AI computing system with 350,000 H100 processors, estimated at over \$10 billion. As Martens notes, *"this barrier to entry is completely beyond the reach of public funding; only the largest companies can aspire to it"*.⁴

For European actors, this asymmetry translates concretely. The cost of training a model comparable to GPT-4 is estimated at approximately \$100 million in the United States (with direct GPU access, cheap energy, amortization over a massive installed base). In Europe, the same training would cost 2 to 5 times more, due to: (i) indirect compute access via US cloud (hyperscaler margins), (ii) energy costs 2 to 3 times higher (Chapter III), and (iii) the absence of comparable economies of scale. The FLOP cost gap (CACI metric M2) is estimated between 2.4× and 3.6×.⁵

4.1.3 Forced Co-opetition: European Startups and US Big Tech

Exponential costs create a mechanism of structural dependence. As analyzed by Martens (Bruegel), AI startups — including European ones — are forced to cooperate with American Big Tech to access computing infrastructure and distribution channels. This *co-opetition* (forced cooperation with a competitor) places startups in a subordinate position: they depend on US platforms to train their models, then find themselves in direct competition with the proprietary models of those same platforms. The UK Competition and Markets Authority (CMA, 2024) and the European Commission have identified these agreements as potentially anti-competitive.⁶ Azoulay et al. (2024), cited by Martens, argue that resource sharing is the only path to avoid concentration, unless a "fox" actor uses openness as a competitive strategy — which Meta has partially done with LLaMA.

4.2 Cloud Concentration as a Vector of Geopolitical Dependence

4.2.1 The European Cloud Market: 70% Under US Control

The second mechanism is the concentration of the European cloud market. Synergy Research Group (July 2025) establishes that the European cloud infrastructure services market reached **€61 billion in 2024** (multiplied by six since 2017) and should exceed €75 billion in 2025 (+24%). **Amazon (AWS), Microsoft (Azure), and Google (GCP) account for 70% of this market.** The share of European providers dropped from 29% in 2017 to 15% in 2022, where it has stagnated since.⁷

Among European providers, SAP and Deutsche Telekom lead with approximately 2% market share each, followed by OVHcloud, Telecom Italia, and Orange. John Dinsdale, Synergy's chief analyst, observes that *"US providers invest €10 billion per quarter in European capex"*, and that the three hyperscalers now operate more than 140 hyperscale data centers in Europe.⁸ The fastest growth (140–160%) concerns services specifically related to generative AI (GPU-as-a-Service, GenAI PaaS), a segment where European players are virtually absent.

Table 8. European cloud market: US hyperscaler domination. Source: Synergy Research Group (July 2025).

Indicator	2017	2022	2024	Trend
EU Cloud Market (€ Bn)	~10	~35	61	×6 in 7 years
AWS+Azure+GCP Share	~50%	~67%	70%	Growing
EU Provider Share	29%	15%	15%	Stable (floor)
US Capex in Europe	n.a.	n.a.	~€40 Bn/yr	€10 Bn/quarter
Hyperscale DCs in EU	~40	~80	>140	+75% in 2 years

4.2.2 From Commercial Lock-in to Geopolitical Lock-in

This concentration goes beyond a simple commercial issue. The concept of *geopolitical vendor lock-in* (defined in Chapter I) describes a situation where technical dependence on a foreign provider is compounded by a regulatory risk linked to the sovereign decisions of that provider's home country. The CLOUD Act (2018) requires American cloud providers to furnish US authorities with data hosted on their servers, including those located in Europe. In May

2025, Microsoft France's chief legal officer acknowledged under oath the inability to guarantee that French citizens' data would never be transmitted to American authorities.⁹

In the AI context, geopolitical lock-in operates at three levels. At the **infrastructure** level (IaaS), AI workloads are deployed on servers physically controlled by entities subject to US jurisdiction. At the **platform** level (PaaS), training and inference frameworks are integrated into proprietary ecosystems (Azure ML, SageMaker, Vertex AI). At the **model** level (MaaS — Model-as-a-Service), access to state-of-the-art models (GPT-4, Claude, Gemini) goes through APIs controlled by US companies, with terms of service that can be modified unilaterally. Section 232 protectionism, by differentiating the cost of hardware access, reinforces all three levels simultaneously.

4.2.3 The Draghi Report and the European Diagnosis

The Draghi Report (September 2024), commissioned by the European Commission, constitutes the most comprehensive institutional recognition of this dependence. Draghi observes that European restrictions on data storage and processing *"create high compliance costs and hinder the creation of large integrated datasets for AI model training"*. He recommends the creation of a European hyperscale cloud infrastructure, the consolidation of European cloud providers, and massive investment in AI, quantum computing, and EuroHPC supercomputers.¹⁰ Martens (Bruegel, 2025) moderates this optimism, however, observing that EuroHPC supercomputers are not designed for generative AI and that upgrading to competitive standards *"exceeds the financial capacity of European budgets"*.¹¹

4.3 Differentiated Productivity: The Abundant Compute Dividend

4.3.1 The Theoretical Macroeconomic Potential

The third transmission channel is AI's impact on labor productivity and its regional asymmetry. The McKinsey Global Institute (May 2024, December 2025) estimates that generative AI could enable Europe to reach an annual productivity growth rate of **3% per year by 2030** in an accelerated adoption scenario with proactive workforce redeployment.¹² This figure is sufficient to close most of the productivity gap with the United States. However, in a slow adoption scenario, productivity growth would be only **0.3%** — close to current levels and insufficient to fund the European social model.

The IMF (Working Paper, March 2025) provides a more granular analysis.¹³ By cross-referencing task exposure indices to AI (Felten et al., Eloundou et al.) with sectoral data, the IMF estimates total factor productivity (TFP) gains *significantly higher in advanced economies than in middle-income countries*, due to their greater share of value added in high AI-exposure sectors (financial services, consulting, technology) and their higher wage levels, which further justify automation. Underlying microeconomic studies (randomized controlled trials) show gains ranging from 14% for low-skilled tasks to over 50% for software engineers.

4.3.2 The Gap Conditioned by Compute Access

Yet, the realization of these productivity gains is conditioned by compute access. This is where the asymmetry diagnosed in Chapter III produces its effects. The French AI Commission (2024) estimates AI's potential impact on French growth at 1.3 percentage points per year, by analogy with the effects of electricity — but this estimate assumes unconstrained access to

cutting-edge compute. Similarly, McKinsey (December 2025) conditions the optimistic European scenario on "*accelerated adoption*" which implies massive infrastructure investments.¹⁴

The investment gap is considerable. McKinsey (January 2026) estimates that large European companies face an annual deficit of **\$700 billion in R&D and capital expenditure** compared to their American counterparts. In digital technologies alone, US companies invested **€2 trillion more** than European companies over the 2021–2025 period.¹⁵ The annualized gap between January 2024 and September 2025 amounts to €580 billion for corporate investment and €300 billion for startups and scale-ups.

This investment deficit, combined with the compute asymmetry and energy costs, produces a **conditional AI productivity gap**. In our modeling, we distinguish between theoretical potential (identical for US and EU with comparable sectoral structures) and *achievable potential*, which depends on effective compute access. With a CACI(US)/CACI(EU) of ~3.4:1 in Power Mode (Chapter III), the achievable European potential is structurally inferior.

Table 9. AI productivity and competitive advantage: US vs EU. Sources: McKinsey (2024, 2025, 2026), Bruegel/Martens (2024), CACI calibration (author).

Dimension	United States	Europe (EU)	Gap
Potential AI productivity (McKinsey, accel. scen.)	+2.5–3.5%/yr	+2.5–3.0%/yr	Comparable
Achievable AI productivity (under compute constraints)	+2.5–3.0%/yr	+0.8–1.5%/yr	-1.5 to -2 pts
Annual AI investment (corporate + VC)	~\$450 Bn	~€50–70 Bn	~7:1
FLOP cost (\$/TFlop, training)	~0.5	~1.2–1.8	2.4–3.6×
CACI ratio (2024 calibration)	Reference	7–12× inferior	~3.4:1

4.4 Innovation Rent Capture: A Self-Reinforcing Advantage

4.4.1 The First-Mover Advantage Mechanism in AI

The fourth channel is the most structurally significant in the long run. The theory of general purpose technologies (GPT, Bresnahan & Trajtenberg, 1995) predicts that early adopters of a general technology capture *disproportionate innovation rents*, which become self-reinforcing through three mechanisms:

First, **data scale effects**. Companies that deploy AI first accumulate usage data (feedback loops, fine-tuning) that improves the performance of their models. This data is non-rival but exclusive: once captured by an actor, it does not benefit competitors. McKinsey estimates that 75% of generative AI's economic potential is concentrated in four key functions (customer service, software development, marketing, R&D), where usage data is particularly determinative.¹⁶

Second, **platform network effects**. US hyperscalers benefit from a virtuous circle: more clients → more revenue → more infrastructure investment → better offering → more clients. This mechanism explains why, despite discussions on digital sovereignty, European providers'

share stagnates at 15%: the service quality gap (latency, range of services, technical support, integration ecosystem) is too large to be bridged by sovereignty arguments alone.

Third, **talent capture**. US companies offer compensation packages including salaries and equity that attract the best AI researchers and engineers worldwide, including Europeans. Martens notes that qualified AI personnel are scarce and that companies compete fiercely to recruit them, with compensation packages *"difficult for European institutions to match"*. This AI *brain drain* weakens the European CACI's L(r) factor and reinforces the American advantage in human capital.¹⁷

4.4.2 The Risk of Irreversibility

These three mechanisms produce a potentially irreversible cumulative effect. As the OECD analyzes (Martens, 2021) by analogy with previous technological waves (broadband internet, cloud), latecomers in the adoption of a general technology become structurally dependent on foreign platforms, suffer compressed margins, and progressively lose their strategic autonomy. The 2022 semiconductor shortage, which cost the European automotive sector alone €100 billion, illustrates the concrete consequences of unanticipated technological dependence.¹⁸

The trajectory is all the more concerning as the EU is absent from **four of the eight segments of the generative AI value chain** as mapped by McKinsey (2024): AI-design semiconductors (dominated by Nvidia), AI cloud platforms (AWS, Azure, GCP), foundation models (OpenAI, Google, Anthropic, Meta), and AI development tools. It is competitive only in downstream application segments: sectoral applications (SAP, Siemens), industrial integration, and specialized semiconductors (power semiconductors: Infineon, STMicroelectronics, NXP, with ~15% global market share).¹⁹

4.5 Synthesis: The Mechanics of US Competitive Advantage

The four channels identified form a reinforcing system. The compute asymmetry (Chapter III) feeds differentiated training costs (4.1), which strengthen dependence on US cloud (4.2), which constrains achievable productivity (4.3), which slows European investment and reinforces rent capture by American players (4.4), which in turn widens the compute gap. This circle constitutes what Farrell and Newman (2019) would describe as structural *interdependence weaponization*: European dependence on American infrastructure, initially founded on economic efficiency, becomes a lever of geopolitical power when instrumentalized by measures such as Section 232.

Trump's protectionism (Chapter III, phase 4) did not *create* this asymmetry — it pre-existed massively. It **institutionalized** and **legalized** it through a tariff mechanism that differentiates the cost of compute access by nationality. In doing so, it transforms a de facto advantage into a de jure advantage, whose dismantling is politically and legally far more difficult. Chapter V examines how these mechanisms could evolve across four prospective scenarios.

Notes

¹ Martens, B. (2024), "The Tension Between Exploding AI Investment Costs and Slow Productivity Growth", Working Paper 18/2024, Bruegel, pp. 5–8. The time series is calibrated on Epoch AI training cost data for frontier models.

² Cottier, B. et al. (2024), cited in Martens (2024), op. cit. The 10× factor between infrastructure and training costs is a central estimate, subject to variation based on post-training infrastructure utilization.

³ IEA (2025), Energy and AI, Paris, p. 42. The IEA compiles investment announcements from Meta, Amazon, Alphabet, Microsoft and xAI for 2025.

⁴ Martens, B. (2024), "Why Artificial Intelligence Is Creating Fundamental Challenges for Competition Policy", Policy Brief 16/2024, Bruegel. Meta plans a system of 350,000 H100s (at ~\$30,000/unit), i.e. >\$10 Bn in hardware alone.

⁵ Author's estimate based on Bruegel (training costs), Eurostat (industrial electricity tariffs), EIA (US prices), and Epoch AI (GPU performance/price) data. The 2.4–3.6× range depends on access mode (cloud vs on-premise) and PPA negotiations.

⁶ Competition & Markets Authority (2024), "AI Foundation Models — Update Report", London. The European Commission launched a call for contributions on competition in generative AI in 2024.

⁷ Synergy Research Group (July 2025), "European Cloud Providers' Local Market Share Now Holds Steady at 15%". The market is defined as IaaS + PaaS + hosted private cloud.

⁸ Dinsdale, J. (2025), cited in Synergy Research Group, op. cit.: "US cloud providers continue to invest €10 billion every quarter in European capex, most of which comes from the big three."

⁹ Data Center Dynamics (July 2025), "European Cloud Providers Hold 15% of Local Market Share". The article reports the testimony of Anton Carniaux, Microsoft France's chief legal officer, before a parliamentary committee.

¹⁰ Draghi, M. (September 2024), The Future of European Competitiveness, report commissioned by the European Commission. Cited in Martens, B. (2025), "Catch-Up with the US or Prosper Below the Tech Frontier? An EU Artificial Intelligence Strategy", Bruegel Policy Brief.

¹¹ Martens, B. (2025), op. cit. The author notes that the EuroHPC approach, centered on hardware, has characterized European digital policies for decades and is "particularly poorly oriented" regarding AI.

¹² McKinsey Global Institute (May 2024), "A New Future of Work: The Race to Deploy AI and Raise Skills in Europe and Beyond". The 3% figure assumes an accelerated adoption scenario with full workforce redeployment.

¹³ IMF (March 2025), "Artificial Intelligence and Productivity in Europe", Working Paper WP/25/067. The analysis uses Felten et al. (2021) and Eloundou et al. (2024) exposure indices.

¹⁴ McKinsey (December 2025), "Accelerating Europe's AI Adoption: The Role of Sovereign AI Capabilities". The report estimates European labor productivity growth at 0.2% on average over 2020–2025, versus a potential of 3.1% with full AI adoption.

¹⁵ McKinsey (January 2026), "Transforming Europe: Bold Moves to Lift a Continent". The \$700 Bn/year gap is calculated on the 150 largest technology companies (US vs EU) in capex + R&D (S&P Capital IQ data).

¹⁶ McKinsey Global Institute (2024), "The Economic Potential of Generative AI: The Next Productivity Frontier". The four functions (customer service, software development, marketing/sales, R&D) represent ~75% of the identified economic potential across 63 use cases.

¹⁷ Martens, B. (2024), Working Paper 18/2024, op. cit. Personnel costs (salaries + equity) are often the most significant component of AI model development costs.

¹⁸ McKinsey (October 2024), "Time to Place Our Bets: Europe's AI Opportunity". The €100 Bn GDP loss figure for the European automotive industry comes from Allianz (September 2022) and OICA.

¹⁹ McKinsey (October 2024), op. cit. European semiconductors (Infineon, STMicro, NXP) represent ~15% of the global integrated design-manufacturing market (Omdia/Informa data, 2024), but primarily in power and automotive segments, not AI GPU/ASIC.

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